Trial SET #2 (math) :



- Don't use calculator and any other materials, including formulae and tables;
- You have only **60** minutes to complete the test;
- Try to do your best and Good Luck!!!

You should be familiar with:

- word problems;
- angles in triangles;
- number properties;
- combinatorics;
- integrals;
- trigonometry;
- arithmetic sequences;
- roots;
- areas in quadrilaterals;
- circles;
- logarithms;
- derivatives;

Name of the student _____

1. If k is an integer, and $\frac{35^2-1}{k}$ is an integer, then k could be each of the following, EXCEPT

A. 8 B. 9 C. 12 D. 16 E. 17

2. The surface of a certain planet reflects 80 percent of the light that strikes it. The clouds around the planet then absorb 40 percent of the reflected light. What percent of the light that strikes the planet is reflected from the surface and passes through the clouds without being absorbed?

A. 32%
B. 40%
C. 48%
D. 60%
E. 88%

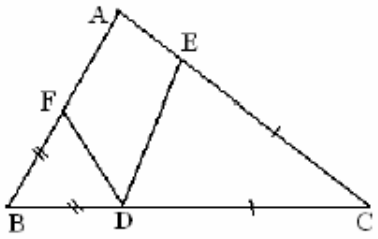
3. Shown below is a correct problem in addition, with R and S representing different digits.

$$\begin{array}{r} 7R \\ RS \\ RR \\ \hline 117 \end{array}$$

What is the value of S ?

A. 1 B. 2 C. 3 D. 4 E. 5

4. In the given figure $BD = BF$, $DC = EC$, $m\angle FDE = 50^0$ then find $m\angle BAC$.



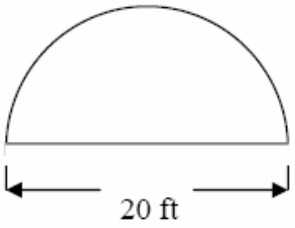
A. 50° B. 60° C. 70° D. 80° E. 90°

5. $\sin^2\frac{\pi}{8}+\cos^2\frac{3\pi}{8}+\sin^2\frac{5\pi}{8}+\cos^2\frac{7\pi}{8}=?$

A. 1 B. 2 C. 3 D. -1 E. 4

6. A particular stock is valued at $50 per share. If the value increases 25 percent and then decreases 20 percent, what is the value of the stock per share after the decrease?

A. $55.00 B. $52.50 C. $50.00 D. $47.50 E. $45.00



7. The figure above shows the dimensions of a semicircular cross section of a one-way tunnel. The single traffic lane is 12 feet wide and is equidistant from the sides of the tunnel. If vehicles must clear the top of the tunnel by at least ½ foot when they are inside the traffic lane, what should be the limit on the height of vehicles that are allowed to use the tunnel?

A. 5½ ft
B. 7½ ft
C. 8 ½ ft
D. 9 ½ ft
E. 10 ft

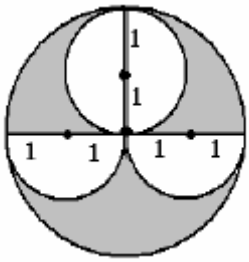
8. If $x=27^{-\frac{1}{3}}$, then find $x^{-\frac{1}{x}}$

A. 27 B. 9 C. 1/3 D. 1/9 E. 81

9. $\begin{cases} \frac{3}{4}(x-1)+\frac{7}{8}<\frac{1}{4}(x-1)+\frac{5}{2}, \\ \frac{x}{4}-\frac{2x-3}{3}<2; \end{cases}$

A. $-2<x<17/8$ B. $-12/7<x<17/8$ C. $12/5<x<17/4$
D. $-1<x<13/4$ E. $-12/5<x<17/4$

10. Two circles and two semicircles are placed as in the given figure. Find the shaded areas.



A) π B) 2π C) 3π D) 4π E) 5π

11. Eight years from now, Carrie's age will be twice Ben's age. If Carrie's current age is C and Ben's current age is B, which of the following represents Carrie's current age?

A. B - 8 B. B + 4 C. B + 8 D. 2B + 8 E. 3B

12. Find the smallest period of the function $y = \cos^2(x)$

A. π B. $\frac{\pi}{2}$ C. $\frac{\pi^2}{2}$ D. $\frac{\pi}{4}$ E. 4π

13. In the sequence 12, 24, 72, 264…, where 12 is the first term, which of the following could denote the *n*th term?

A. $12\times n$; B. n^{12}; C. 12^n; D. $4^{(n+1)}-n$; E. 4^n+8;

14. Simba borrowed $12,000 from his brothers so he can buy a new sports car. If Simba returns 4.5% of that amount every 2 weeks, after how many months Simba wouldn't owe his brothers any more money?

A. 8; B. 12; C. 15; D. 18; E. 20;

15. Two pens, four rulers and two notebooks cost $21. Five pens, four rulers and three notebooks cost $17. How much more expensive is a ruler than a pen?

A. $7.25 B. 6$ C. $3.25 D. $8.00 E. More information is needed

16. What is the solution set for $\frac{e^x - e^{-x}}{e^x + e^{-x}} = \frac{1}{2}$?

A. $\frac{1}{2}\ln 3, -\frac{1}{2}\ln 3$ B. $\ln\sqrt{3}$ C. $2\ln 3$ D. $\frac{1}{2}\ln\sqrt{3}$ E. $-\ln\sqrt{3}$

17. One integer is chosen at random from the numbers 1, 2, 3, ..., 50. What is the probability that the chosen number is divisible by 6 or by 8?

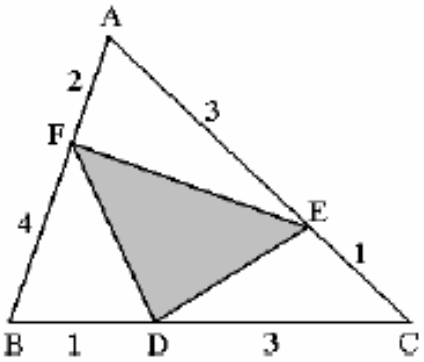
A. 0,2 B. 0,22 C. 0,24 D. 0,26 E. 0,28

18. $\log_6 3=a \Rightarrow \log_6 2=?$

A. a-2 B. 1-a C. 1+a D. 1-2a E. 1/a

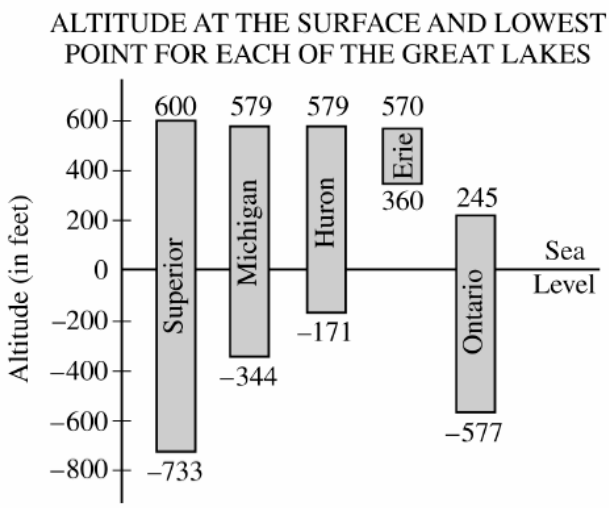
19. If x and y are positive integers ($x > y$), what is the units' digit of $(10^x - 9^y)^2$?

A. 9; B. 7; C. 5; D. 3; E. 1;



20. In the given triangle EC=BD=1, DC=AE=3, AF=2, FB=4, Find the ratio of the area of triangular region DEF to the are of trianglular region ABC.

A. $\frac{19}{128}$; B. $\frac{19}{48}$; C. $\frac{25}{48}$; D. $\frac{25}{64}$; E. $\frac{1}{2}$;



21. The depth of a lake is the difference between the altitude at the surface and at the lowest point of the lake. If the five lakes in the graph above were listed in order from the greatest depth to the least depth, which lake would be third in the list?
(A) Erie
(B) Huron
(C) Michigan
(D) Ontario
(E) Superior

22. The sum of the positive odd integers less than 100 is subtracted from the sum of the positive even integers less than or equal to 100. What is the resulting difference?

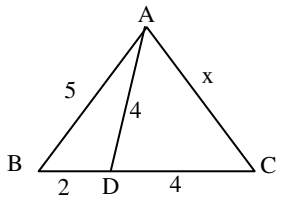
A. 50; B. 49; C. 2500; D. 52; E. 1;

23. A credit card number has 6 digits (between 1 to 9). The first two digits are 12 in that order, the third digit is bigger than 6, the forth one can be equally divided by 3 and the fifth digit is 3 times bigger than the sixth one. How many credit cards can be made?
(a) 27
(b) 36
(c) 72
(d) 112
(e) 422

24. $y = \ln(2x+1)^2 \Rightarrow \dfrac{dy}{dx} = ?$

A) $\dfrac{1}{2x+1}$ B) $\dfrac{2}{(2x+1)^2}$ C) $\dfrac{4}{(2x+1)^2}$ D) $\dfrac{4}{2x+1}$ E) $\dfrac{2}{2x+1}$

25. $|AB|=5$ cm, $|BD|=2$cm, $|AD|=|DC|=4$cm
Find $|AC| = x$.



A. $\sqrt{19}$ B. $2\sqrt{5}$
C. $\sqrt{21}$ D. $\sqrt{22}$
E. $\sqrt{23}$

26. $f(x) = (2+x)^2\left(2-x^2\right)^2$ then $f'(-1) = ?$

A.–12 B. –9 C. 9 D. –6 E. 6

27. y = xctgx is given. f′(x)=?

A. $-\dfrac{x}{\sin^2 x}$ B. ctgx C. ctgx $-\dfrac{x}{\sin^2 x}$

D. $1+\dfrac{x}{\sin^2 x}$ E. $1-\dfrac{x}{\sin^2 x}$

28. Find the area of the region bounded by the curves $x = y^3$ and $y = x^3$.

A. $\frac{1}{2}$ B. 1 C. $\frac{3}{2}$ D. 2 E. 3

29. Solve $\tan x + \cos 2x = 2$

A) $k\pi$ B) $\frac{\pi}{3} + 3k\pi$ C) $\frac{\pi}{2} + 2k\pi$ D) None of these E) $\frac{\pi}{2} + k\pi$

30. If $\frac{r+t}{r-t} = \frac{5}{2}$, what is the value of $\frac{r}{t}$?

A. $-\frac{7}{3}$; B. -1; C. $\frac{3}{7}$; D. 1; E. $\frac{7}{3}$;